ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) Resolution of the ADC = $V_{range}/2^n$ = 5 V / 4096 = 1.22×10^{-3} V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau} B \, d\tau = A^{-1}(F - I)B$ ($A \neq 0$).

Here A = −2, B = 1, T = 0.1: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 2

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\, x_2 + mg\ell \sin x_1]$.

At equilibrium $x_2 = 0$ and $mg\ell \sin \theta_e = -\tau_0$, so $|\tau_0| \leq mg\ell = 2.4525$ N·m.

(b) Jacobian at $\theta_e = 60°$: A = [0, 1; $-(mg\ell/J)\cos\theta_e$, −b/J].

$mg\ell/J = 2.4525/0.02 = 122.625$; cos(60) = −0.9524, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At steady state $\dot{\theta} = 0$ and $\ddot{\theta} = 0$.

A PD controller is stabel and its job is to drive $(\theta_d - \theta)$ to zero, so $\theta_{ss} = \theta_d = 90°$. The gravity term only affects the transient. Hence the stedy-state error is **0°** and the joint does come to rest at θ_d.

(d) With the compensation term, at rest $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss}$, so $\theta_{ss} = \theta_d$ and the steady-state error is **0**.

Question 4

(a) Substitutting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\, r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving s = −0.586 and s = −3.414. DC gain = $bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) The actuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) Substituting u into the plant and using $\dot{\xi} = -\theta$ (r = 0): characterstic polynomial $s^3 + (a + k_2)s^2 + (c + k_1)s + K_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; **$K_I = 32$**.

(d) For a constant r, in stedy state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the acuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrator, conditional integration or back-calculation).